

For questions 1 through 4, use the table showing Lines B and M .

Line B	Line M
$B(x) = \frac{3}{5}x - 4$	

1. Which linear function has the steeper slope?

Line M

2. Which line has the larger x -intercept?

Line B

3. Name all the key feature(s) that are the same for both linear functions.

Both linear functions have the same domain and range. They are also both increasing.

4. Create a linear function that has the same y -intercept as Line B .

Answers will vary. $C(x) = \frac{1}{4}x - 4$

5. Complete the statements below to describe the end behavior of Line B .

- a. As $x \rightarrow +\infty$, $y \rightarrow$

☐ $-\infty$.
☒ ∞ .

- b. As $x \rightarrow -\infty$, $y \rightarrow$

☒ $-\infty$.
☐ ∞ .

For questions 6 through 10, use the table showing Lines G and R .

Line G	Line R										
x -int: -6 A line passes through the points $(-6, 0)$ and $(4, 15)$. Slope: $\frac{15-0}{4-(-6)} = \frac{15}{10} = \frac{3}{2}$ y -int: $0 = \frac{3}{2}(-6) + b$ $0 = -9 + b$ $9 = b$	Slope: $\frac{5-4}{0-(-3)} = \frac{1}{3}$ y -int: 5 <table border="1"> <thead> <tr> <th>x</th><th>y</th></tr> </thead> <tbody> <tr> <td>-3</td><td>4</td></tr> <tr> <td>0</td><td>5</td></tr> <tr> <td>3</td><td>6</td></tr> <tr> <td>6</td><td>7</td></tr> </tbody> </table>	x	y	-3	4	0	5	3	6	6	7
x	y										
-3	4										
0	5										
3	6										
6	7										

6. Which linear function has the smaller slope?

Line R

$$\begin{aligned}
 x\text{-int: } 0 &= \frac{1}{3}x + 5 \\
 -5 &= \frac{1}{3}x \\
 -15 &= x
 \end{aligned}$$

7. Which linear function has the larger y -intercept?

Line G

8. Name the key feature(s) that are different for the functions.

Line G and Line R have different x - and y -intercepts and slopes.

9. Create a linear function that has the same slope as Line R .

Answers will vary. $D(x) = \frac{1}{3}x - 6$

10. Consider the end behavior for the two functions.

- a. Complete the statement.

Lines G and R have

- ☒ the same end behavior.
☐ different end behaviors.

- b. Write the end behavior of the functions.

As $x \rightarrow \infty$, $y \rightarrow \infty$.

As $x \rightarrow -\infty$, $y \rightarrow -\infty$.